

ZAHRA KHATTI

Bethlehem, PA

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Summary

ML researcher bringing optimization rigor to modern deep-learning systems. First-author of a peer-reviewed paper (IOS 2026) on enforcing *exact* fairness targets in supervised learning through constrained optimization and smooth nonconvex surrogates—guarantees in place of penalty tuning, with <5% accuracy cost. Proficient in PyTorch (GPU/CUDA), HuggingFace Transformers, and PEFT/LoRA, with deep grounding in gradient-based and constrained optimization and end-to-end experiment pipelines.

Education

Lehigh University

Ph.D. in Industrial and Systems Engineering | GPA: 3.71/4.0

Bethlehem, PA

- Advisors: Prof. Frank E. Curtis, Prof. Daniel P. Robinson
- **Focus:** Machine Learning, Fairness in AI, Constrained ML, Stochastic Optimization
- **Coursework:** Statistical & Machine Learning, Optimization Methods for ML, Convex Analysis, Continuous Optimization, Probability & Stochastic Processes

Sharif University of Technology

M.Sc. in Industrial Engineering | Full Scholarship, “Exceptionally Talented” Designation

2019 – 2022

Tehran, Iran

- Thesis: Customer Segmentation and Migration Pattern Modeling Based on Loyalty
- **Coursework:** Econometrics, Stochastic Processes, Stochastic Programming, Dynamic Systems

K.N. Toosi University of Technology

B.Sc. in Industrial Engineering | Ranked 3rd of 42, Full Scholarship

2015 – 2019

Tehran, Iran

Technical Skills

- **Languages & Tools:** Python, SQL, R, MATLAB, Git/GitHub, Linux, LaTeX, Bash
- **ML / Deep Learning:** PyTorch (GPU/CUDA, distributed), HuggingFace Transformers, PEFT/LoRA, Scikit-learn, NumPy, SciPy, Pandas; Neural Networks, Transformers, Attention, Tokenization (BPE), Fine-tuning, CNNs, Ensemble Methods (XGBoost, Random Forest)
- **Optimization & Solvers:** Gurobi, AMPL, PuLP, CUTEst; Constrained Optimization, Stochastic Optimization, Gradient-based Algorithms, SQP, Interior-Point Methods
- **Data Science & Analytics:** Statistical Modeling, Predictive Analytics, A/B Testing, Causal Inference, Feature Engineering, Hypothesis Testing
- **Software Engineering & Infrastructure:** pytest, Docker, virtualenv/conda, reproducible research workflows, HPC (SLURM), AWS basics

Research & Professional Experience

Graduate Research Assistant — Machine Learning & Optimization

Aug. 2023 – Present

Lehigh University, Dept. of Industrial and Systems Engineering

Bethlehem, PA

- **Fair Supervised Learning with Constrained Optimization** (IOS 2026; INFORMS 2025; NeurIPS 2025 Workshop)
 - Designed a constrained-optimization framework that enforces *exact* fairness targets in supervised learning ($\delta=0.8 \rightarrow \text{actual} \approx 0.8$) with <5% accuracy loss—guarantees in place of penalty-weight tuning.
 - Engineered smooth nonconvex surrogates for discontinuous fairness measures (demographic parity, equal opportunity), enabling gradient-based training; built the full PyTorch pipeline—preprocessing, training loops, fairness evaluation, automated experiment orchestration—across Dutch, Law School, and ACSIncome.
 - First-author, peer-reviewed: IOS 2026, arXiv:2505.15788.
- **Low-Order Hessian-Imitation Optimizer for Neural-Network Training**
 - Co-developed an optimizer that approximates second-order curvature via an auxiliary loss at Adam-level cost, with a convergence guarantee and benchmark wins over Adam. Co-author: arXiv:2605.06945.
- **Adaptive Sampling Interior-Point Method (AS-IPM) for Stochastic Optimization**
 - Developed an adaptive-sampling interior-point solver—an optimization engine for training under noisy, data-driven objectives—combining retrospective approximation with barrier methods.
 - Implemented in Python/SciPy with adaptive sample-size control; validated on CUTEst problems at 4,000+ variables, with convergence guarantees for feasibility and stationarity.

Graduate Research — Customer Analytics & Predictive Modeling

2019 – 2022

Sharif University of Technology (Master's Thesis)

Tehran, Iran

- Built a dynamic customer-segmentation system on two years of transaction data using K-Means clustering on RFM (Recency, Frequency, Monetary) features across consecutive time periods.
- Implemented PrefixSpan sequential pattern mining to detect customer migration patterns; separated high-potential churning customers from permanently lost ones with actionable retention strategies.

Selected Projects

- Constrained LoRA Fine-Tuning for Bias-Controlled LLMs

2026
- In Progress*
- Fine-tuning Language Models with LoRA under a hard, user-set fairness constraint, so a target bias level is enforced during training rather than approximated with hand-tuned penalties that can’t guarantee it.
 - Built the training stack: a custom constrained optimizer wrapping the HuggingFace PEFT **Trainer**, with BBQ/BOLD bias benchmarks and task accuracy tracked in the same loop.
 - Core question: whether the exact-tolerance guarantee proven on tabular data holds at LLM scale without sacrificing accuracy.

Publications

- Z. Khatti, D. P. Robinson, F. E. Curtis. **Fair Supervised Learning Through Constraints on Smooth Nonconvex Unfairness-Measure Surrogates**. Refereed Proceedings of the 2026 INFORMS Optimization Society Conference (IOS 2026), Atlanta, GA. [arXiv:2505.15788]
- Y. Zhu, L. Guo, Z. Khatti, X. Qu, C.-Y. Wu, L. Zebiane, F. E. Curtis. **Low-Order Explicit Hessian Imitation Method for Large-Scale Supervised Machine Learning**. Preprint, arXiv:2605.06945, 2026.

Selected Presentations

- **IOS 2026**, Atlanta, GA — contributed talk (fairness in ML).
- **INFORMS Annual Meeting 2025**, Atlanta, GA — contributed presentation.
- **NeurIPS 2025 COML Workshop** — contributed talk and poster.

Leadership & Service

- Co-Chair — MOPTA 2026 Conference & Optimization Workshops (GradOpt, ColOpt)

2025 – 2026
- Lehigh University*
- Bethlehem, PA*
- Co-organizing MOPTA 2026, a major international optimization conference at Lehigh; coordinating invited speakers, session scheduling, and logistics.
 - Co-chairing the GradOpt graduate workshop and the ColOpt junior-faculty colloquium in optimization and data science.
- Executive Board Member, Scientific Association

2016 – 2017
- K.N. Toosi University of Technology*
- Tehran, Iran*

Honors & Awards

- **“Exceptionally Talented” Student** designation, Department of Industrial Engineering, Sharif University of Technology
- **Full Master’s Scholarship**, Sharif University of Technology
- **Ranked 3rd of 42**, B.Sc. Industrial Engineering, K.N. Toosi University of Technology
- **Full Bachelor’s Scholarship**, K.N. Toosi University of Technology